EE3001: Solid State Devices
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Assignment #8

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10}cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$3.2 \times 10^{19}cm^{-3}$, N_v=$1.8 \times 10^{19}cm^{-3}$, $\mu_n = 1350cm^2/V.s$, $\mu_p = 450cm^2/V.s$; Permittivity for vacuum: ϵ_0=8.85×10^{-14}F/cm, relative permittivity: $\epsilon_{r,Si}$=11.8, ϵ_{r,SiO_2}=3.9; pn diode dopants are completely ionized.

1. (2 points) At a particular wavelength of incident radiation, the absorption coefficients of Si and GaAs are 4000 per cm and 30,000 per cm, respectively. What are the thickness of Si and GaAs necessary to absorb 80% of the incident radiation?

2. (5 points) A 5 μm thick Si sample is illuminated with monochromatic light having energy of 2 eV. The incident power is 10 mW.

   (a) What is the total energy absorbed in the sample per second?

   (b) How much energy per second is dissipated as heat?

   (c) How many EHPs are generated in the sample per second?

3. (5 points) A solar cell has an area of 1 cm $\times$ 1 cm. The cell has an n^+p structure, where the doping concentration of the p-region is 10^{15} $/cm^3$. Considering μ_n = 1350 cm^2/V-sec and τ_n = 1 μs in the p-region, and I_L = 40 mA, calculate I_{sc}, V_{oc}, maximum power output (P_m) and the Fill Factor (FF). Estimate the incident optical power if the efficiency of the solar cell is 15 %.

4. (15 points) This problem will help us understand how we can derive maximum power from a solar cell. Also, it is important to adjust the load in order to obtain maximum power at different conditions, viz. sunny or cloudy; morning, noon or evening.

   Background information: To generate power from a p-n solar cell, a load resistor (R) can be connected across it. The current (I) flowing through the solar cell also flows through R. The voltage across the resistor V=-IR also appears across the diode, which is forward biased by the same voltage (V). Hence, the current through the solar cell consists of two components: (i) forward biased current from p to n, and (ii) light generated current (I_L) from n to p. Total current is given as

$$I = I_0(exp(V/V_T) - 1) - I_L \quad (1)$$

and the power generated by the solar cell (and dissipated across R) can be given as

$$P = -IV \quad (2)$$

Problem: The reverse saturation current (I_0) in a p-n junction solar cell is 100 pA.

(a) The solar cell is taken out in the sun. The incident light results in a light generated short-circuit current ($I_{sc} = I_L$) of 40 mA through the solar cell. For this condition of sunlight,

(i) what would be the open circuit voltage (V_{oc}) across the two terminals of the p-n diode?

(ii) what would be the maximum possible power (P_m) which can be delivered by the solar cell to a load resistor connected across it?

(iii) what is the fill factor (FF=$\frac{P_m}{V_{oc}I_{sc}}$) of the solar cell?

(iv) what is the value of R (let's call it $R_{m40} = -\frac{V_m}{I_m}$) required to obtain maximum power across the load?

(b) After some time it became cloudy and I_L is reduced to 10 mA. For this condition of sunlight,

(i) what is the open circuit voltage (V_{oc})?

(ii) what would be the maximum possible power (P_m) which can be delivered by the solar cell to a load resistor connected across it?

(iii) what is the fill factor (FF) of the solar cell?

(iv) what is the value of R (let's call it R_{m10}) required to obtain maximum power across the load?

(c) Suppose we did not change the load resistance and we had $R = R_{m40}$ when $I_L = 10mA$.

(i) Draw the I-V characteristics of the solar cell for the two conditions, i.e., for $I_L = 40$ mA and $I_L = 10$ mA. Draw a straight line (called the load line) corresponding to $V = -IR_{m40}$. The operating point has to be the point of intersection of the load line and the I-V characteristics. Show that with $I_L = 10$ mA and $R = R_{m40}$, $I = -I_L$. This can be verified by calculating V if $I = -I_L$ and substituting this value of V in eq. (1).

(ii) What is the power delivered by the solar cell?

(iii) What is the fraction of power that is available from the solar cell but is wasted by not changing the load resistance from R_{m40} to R_{m10}?